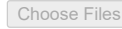


```
# Project Title : Car Price Prediction using Machine Learning
# Internship : CodeAlpha Data Science Internship
# Task : Task 3
# Developed By : Tuba Ousaf

# STEP 1:UPLOAD AND LOAD DATASET
from google.colab import files
uploaded = files.upload()
```

 car data.csv
car data.csv(text/csv) - 16908 bytes, last modified: 7/9/2026 - 100% done
 Saving car data.csv to car data.csv

```
import pandas as pd

df = pd.read_csv("car data.csv")

print("Dataset Loaded Successfully!")
```

Dataset Loaded Successfully!

```
# STEP 2: IMPORT LIBRARIES
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LinearRegression

from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)
```

```
# STEP 3: EXPLORE THE DATASET

print("\nDataset Shape:")
print(df.shape)

print("\nDataset Information:")
df.info()

print("\nStatistical Summary:")
print(df.describe())

print("\nMissing Values:")
print(df.isnull().sum())

# STEP 4: DATA PREPROCESSING
# Create a copy of dataset
car_data = df.copy()

# Remove Car_Name because it is not useful for prediction
car_data = car_data.drop("Car_Name", axis=1)

# Convert categorical columns into numeric values
encoder = LabelEncoder()

car_data["Fuel_Type"] = encoder.fit_transform(car_data["Fuel_Type"])
car_data["Selling_type"] = encoder.fit_transform(car_data["Selling_type"])
car_data["Transmission"] = encoder.fit_transform(car_data["Transmission"])

# STEP 5: SELECT FEATURES AND TARGET

X = car_data.drop("Selling_Price", axis=1)

y = car_data["Selling_Price"]

print("\nFeatures Shape:", X.shape)
print("Target Shape:", y.shape)
```

```

# STEP 6: SPLIT THE DATASET

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

print("\nTraining Samples:", X_train.shape[0])
print("Testing Samples:", X_test.shape[0])

# STEP 7: TRAIN THE MACHINE LEARNING MODEL

model = LinearRegression()

model.fit(X_train, y_train)

print("\nModel Training Completed Successfully!")

# STEP 8: MAKE PREDICTIONS

y_pred = model.predict(X_test)

# STEP 9: EVALUATE THE MODEL

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("\n===== MODEL PERFORMANCE =====")

print(f"Mean Absolute Error : {mae:.2f}")
print(f"Mean Squared Error : {mse:.2f}")
print(f"Root Mean Squared Error : {rmse:.2f}")
print(f"R2 Score : {r2:.2f}")

# STEP 10: VISUALIZE ACTUAL VS PREDICTED VALUES
plt.figure(figsize=(8,6))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Selling Price")
plt.ylabel("Predicted Selling Price")
plt.title("Actual vs Predicted Selling Price")
plt.plot(
    [y_test.min(), y_test.max()],
    [y_test.min(), y_test.max()],
    color="red"
)
plt.grid(True)
plt.show()

#STEP 11: PREDICT SELLING PRICE OF A TEST CAR
# Select the first car from the test dataset
sample_car = X_test.iloc[[0]]

# Predict the selling price
predicted_price = model.predict(sample_car)

# Get the actual selling price
actual_price = y_test.iloc[0]

print("\n===== CAR PRICE PREDICTION =====")
print(f"Predicted Selling Price : {predicted_price[0]:.2f} Lakhs")
print(f"Actual Selling Price : {actual_price:.2f} Lakhs")

# STEP 12: CONCLUSION

print("\nCar Price Prediction Completed Successfully!")

```

